

RITVIJ NARAM

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MACHINE LEARNING SOFTWARE DEVELOPER | GENERATIVE AI

PROFESSIONAL SUMMARY

Information Systems graduate student and published researcher with 3+ years of experience in data analytics, SQL-based data engineering, ETL pipelines, and cloud platforms. Experienced in building data pipelines, dashboards, and AI-driven automation solutions, improving reporting efficiency by up to 97% and enabling data-driven decision-making. Hands-on ML/GenAI work includes TensorFlow/Keras, scikit-learn, LangChain/RAG, Gemini API, Docker, and Git/GitHub CI/CD.

EDUCATION

Master of Science, Information Systems | Northeastern University | Toronto, ON | December 2026

Bachelor of Engineering (BE), Computer Science | University of Pune | Pune, India | May 2021

CERTIFICATIONS

Certified Airtable Builder , Airtable, Canada | May 2025

CORE COMPETENCIES

Machine Learning & GenAI: Python | TensorFlow/Keras | scikit-learn | CNNs | LangChain | LLMs | RAG | Gemini API | Prompt Engineering | Embeddings | Model Evaluation

MLOps & Cloud Platforms: Docker | GitHub Actions CI/CD | Azure | AWS | GCP, Vertex AI, BigQuery & Kubernetes (current hands-on learning) | Model Monitoring

Data Analytics & Engineering: SQL | Python | Data Modeling | ETL Pipeline Development | Data Warehousing | Data Cleaning | Data Transformation | API Integration

Business Intelligence & Visualization: Tableau | Power BI | Dashboard Development | KPI Reporting | Data Storytelling | Business Performance Analytics

Cloud & Data Platforms: Microsoft Azure | Snowflake | Azure Functions | Cloud Data Pipelines | Serverless Architecture

AI & Automation: AI Chatbots | LangChain | LLM Integration | Process Automation | Workflow Optimization | NLP Applications

Database & Systems: Relational Databases | Database Design | Data Governance | Airtable | SharePoint | Enterprise Data Systems

Interpersonal: Cross-Functional Leadership | Problem Solving | Analytical Thinking | Requirements Gathering | Team Collaboration

Programming & Tools: Python | FastAPI | Git | REST APIs | Microsoft Excel | Advanced Excel Automation | Java | C++

PROFESSIONAL EXPERIENCE

Business Analytics & Systems Associate Co-op | Northeastern University | Toronto, ON | Jul 2025 - Dec 2025

Led end-to-end campus analytics and systems initiatives, translating stakeholder requirements into production-ready solutions supporting thousands of users.

Solely owned and delivered the Scorecard Data Project end to end, from stakeholder requirements and architecture through implementation, validation, documentation, deployment, and handoff; built scheduled Azure Functions ETL to pull Airtable/Asana API data, stage CSV/JSON checkpoints in Azure Blob Storage, and load curated Snowflake tables for Tableau reporting, improving data reliability by 50%.

Secured the Scorecard pipeline with Azure Managed Identity, Key Vault, and environment variables; configured Snowflake warehouse/database/schema access and implemented incremental loads using monthly partitions and data hashes to prevent unnecessary reprocessing.

Benchmarked pipeline performance and documented architecture, recovery procedures, and troubleshooting; standardized Git/GitHub workflows, GitHub Actions CI/CD, and Docker-based deployment practices for repeatable releases.

Current technical upskilling: recreating ML deployment patterns on GCP with BigQuery, Vertex AI, Docker, and Kubernetes, including CI/CD, model monitoring, and container orchestration.

Developed Python/Flask services and REST API integrations while maintaining mentorship-program databases and automation workflows supporting 300+ student and alumni records.

Software Developer | Mechanize Industries | Pune, IN | Sep 2022 - May 2024

Built SQL-based ETL pipelines for weekly/monthly operational reporting, applying reusable validation and transformation logic to reduce manual processing and improve reporting consistency.

Designed Power BI dashboards for 15+ KPIs and modeled OLTP operational data into analytics-ready structures, enabling 20+ stakeholders to monitor performance and investigate trends.

Developed an AWS data-migration proof of concept using S3 for staged monthly data, RDS for transactional workloads, and Redshift for columnar analytics; applied partitioning and incremental-load checks to minimize redundant processing.

Automated legacy Excel reporting with Python and SQL, reducing manual workload by ~30%; used Git-based code review and CI/CD practices to improve release consistency and maintainability.

Partnered with business and technical stakeholders to validate datasets, benchmark pipeline performance, document architecture, and drive issues through production handoff.

Software Developer | Raja Industries | Pune, India | Aug 2021 - Aug 2022

Developed an integrated reporting solution extending legacy operational systems with near-real-time analytics, covering requirements, database design, testing, deployment, and support.

Built Python/SQL ingestion workflows and REST API integrations with incremental processing and data-quality checks to prepare operational data for downstream analytics.

Prototyped AWS analytics architecture using S3 staging, RDS source data, and Redshift reporting models, separating transactional and analytical workloads for scalable querying.

Containerized Python services with Docker and used Git/GitHub CI/CD practices for repeatable builds, testing, and deployments while optimizing complex SQL for reporting and audit requests.

Produced technical documentation and user guides; collaborated with analysts and business users to troubleshoot issues, refine requirements, and improve system adoption.

RESEARCH PUBLICATIONS

Anomaly Detection | Link | July - 2021

Published in the International Research Journal of Modernisation in Engineering Technology and Science.

Artificial Intelligence System for Effective Detection of Animal Intervention in Croplands

| Link | Sept - 2021

Published in the International Journal for Research in Applied Science and Engineering Technology.

Wheeled Robot Development for Obstruction Avoidance Mechanism | Link | Sept - 2021

Published in the International Journal for Research in Applied Science and Engineering Technology.

PROJECTS

Anomaly Detection for Video Surveillance | Python, TensorFlow/Keras, OpenCV, CNN

Built a CNN-based video anomaly-detection pipeline that preprocesses frames with OpenCV/NumPy, performs Keras/TensorFlow inference, and annotates abnormal events in video footage.

AI Chatbot for Student Support (Nora)

Built LLM-based chatbot using LangChain and API integrations for automated student query handling and developed backend using Python and APIs.

Improved response automation, reducing manual query handling and increasing response efficiency.

Implemented retrieval-augmented generation (RAG) for context-aware responses.

Local Eco-Scanner | Kotlin, Jetpack Compose, CameraX, Gemini API | GitHub

Developed a native Android application using Kotlin and Jetpack Compose, integrating CameraX and Gemini multimodal AI to identify household items and generate location-aware recycling guidance.

Implemented camera lifecycle and permission handling, image rotation and downscaling, coroutine-based API requests, Canadian postal-code validation, error states, and private local-profile persistence.